NETWORK LOAD-BALANCER

1.create a vpc 11.0.0.0/16

2.create a subnet

1. public-subnet-1a (10.0.0.0/24)
2. public-subnet-1b (10.0.1.0/24)
3. private-subnet-1a (10.0.2.0/24)
4. private-subnet-1b (10.0.3.0/24)

3.create a internet-gateway & attach to vpc

4.create a route table

1. public-route-table
    1) route --> IGW (0.0.0.0/0)
    2) subnet -> public-subnet 1a & 1b
2. private-route-table
    1) subnet -> private-subnet 1a & 1b

5.create a ec2-instances

1. public-server-1
2. public-server-2

6.access public-server-1

1. apt update
2. apt install apache2 -y
3. vim /var/www/html/index.html
   <h1> These is my public-server-1 </h1>
4. systemctl restart apache2
5. curl localhost

7.access public-server-2

1. apt update
2. apt install apache2 -y
3. vim /var/www/html/index.html
   <h1> These is my public-server-2 </h1>
4. systemctl restart apache2
5. curl localhost

8.create a target-group

1. name -----> tg-nlb
2. choose ---> instances
3. protocol -> TCP

4. vpc ------> my-vpc
5. register -> public-server 1&2
6. create

9.create a network load balancer

1. name -----> nlb
2. vpc --------> my-vpc
3. az ----------> public-subnet 1a&1b
4. forward --> tg-nlb
5. create a network load balancer

10.copy the dns of network-load-balancer

1.dns

{ You get server1 and server2 }

Note:

Path-Based Routing is NOT Supported in NLB

Suppose we create the following pages on both servers:

Public-Server-1
http://<server1-ip>/
This is my public-server-1

http://<server1-ip>/mobile/
This is my mobile page from public-server-1

http://<server1-ip>/laptop/
This is my laptop page from public-server-1

Public-Server-2
http://<server2-ip>/
This is my public-server-2

http://<server2-ip>/mobile/
This is my mobile page from public-server-2

http://<server2-ip>/laptop/
This is my laptop page from public-server-2

we thinks:
http://<nlb-dns>/mobile ---> public-server-1
http://<nlb-dns>/laptop ---> public-server-2

But it does not route traffic in these way

Why?
Network Load Balancer works at Layer 4 (Transport Layer) and only checks:

1. Source IP
2. Destination IP
3. Protocol (TCP/UDP)
4. Port Number

It does NOT inspect:

1. /mobile
2. /laptop
3. /index.html

Therefore, NLB cannot make routing decisions based on URL paths.

1. NLB = Connection-based routing (TCP/UDP)
2. ALB = Path-based routing and Host-based routing (HTTP/HTTPS)
